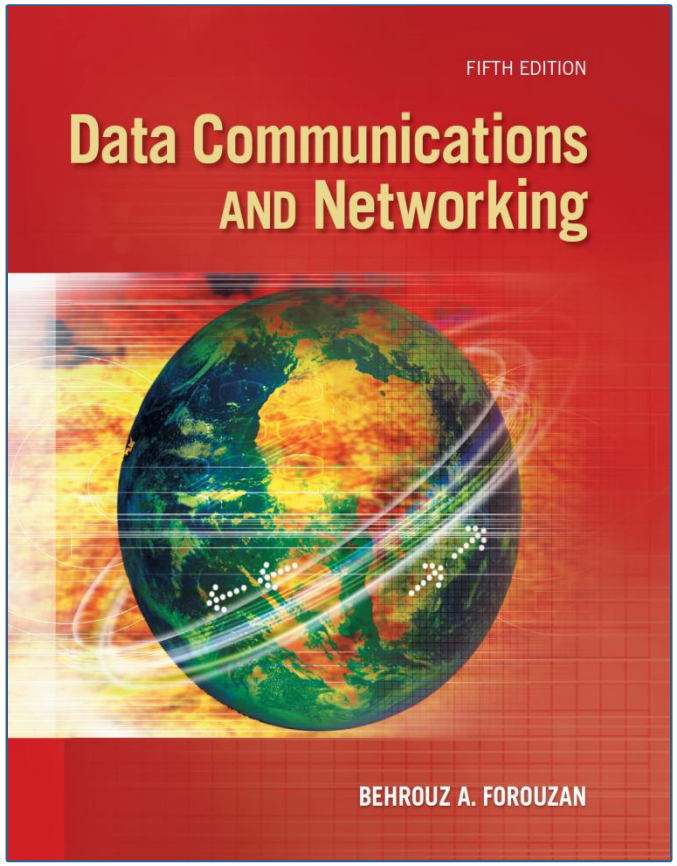


Network Services (Multimedia in the Internet)

Lecturer: Dr. Khaled BinAbdl

Lecture 9

Multimedia in the Internet

Multimedia in the Internet

– **Multimedia services** in the **Internet** can be **divided** into **three** broad **categories**:
  - *streaming **stored** audio/video.*
  - *streaming **live** audio/video.*
  - ***interactive** audio/video.*

– **Streaming** means: a user can **listen** (or **watch**) the file **after** the **downloading** has **started**.

Streaming Stored Audio/Video

- Audio/Video **files** are **compressed** and **stored** on a server.
- A client **downloads** the **files** through the Internet.
- This is sometimes **referred** to as ***on-demand*** *audio/video*.

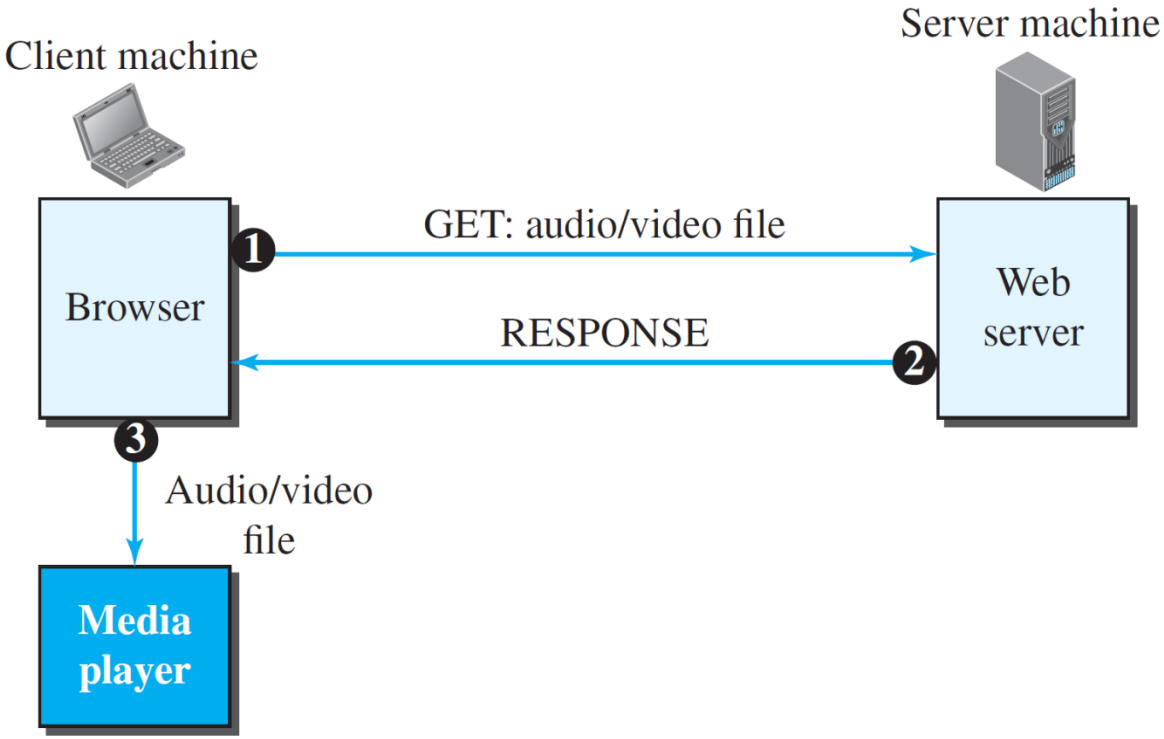
Streaming Stored Audio/Video

– ***Four Approaches:***

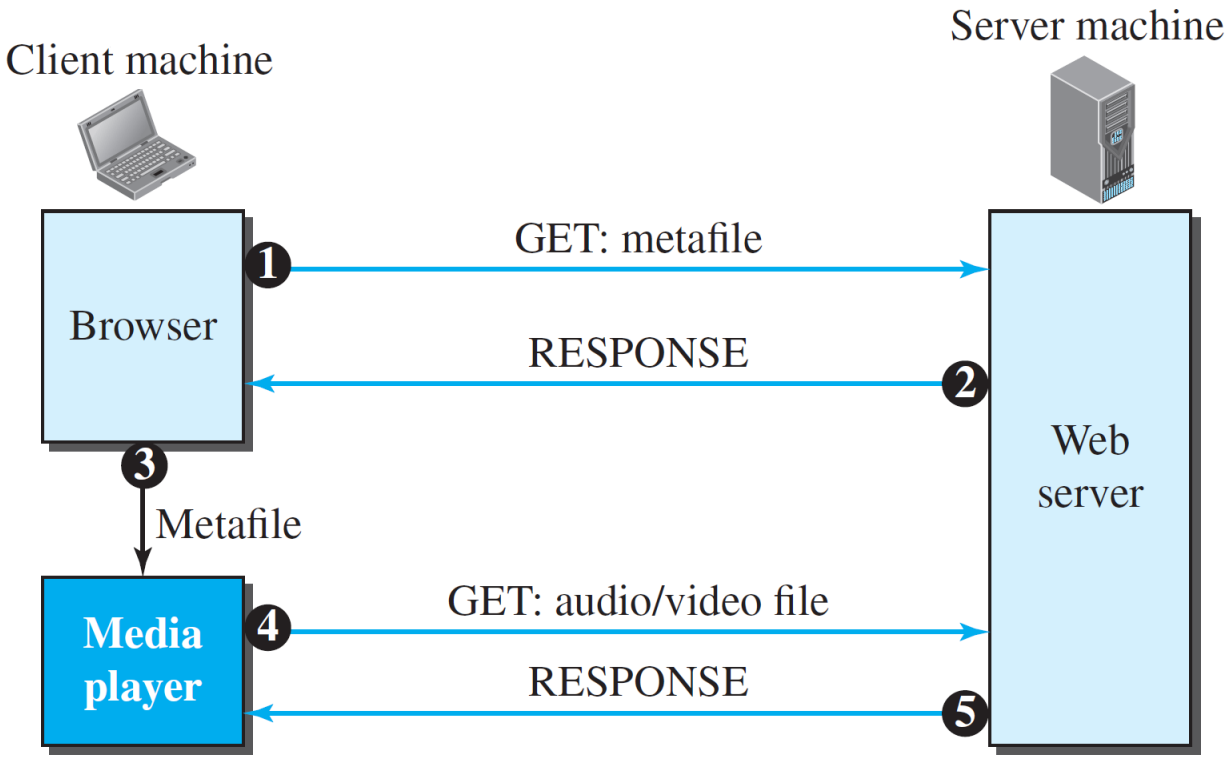
- ***Using a Web Server.***
- ***Using a Web Server with a Metafile.***
- ***Using a Media Server.***
- ***Using a Media Server and RTSP.***

Using a Web Server

– This method does not include streaming, the multimedia file needs to download completely before it can be played.



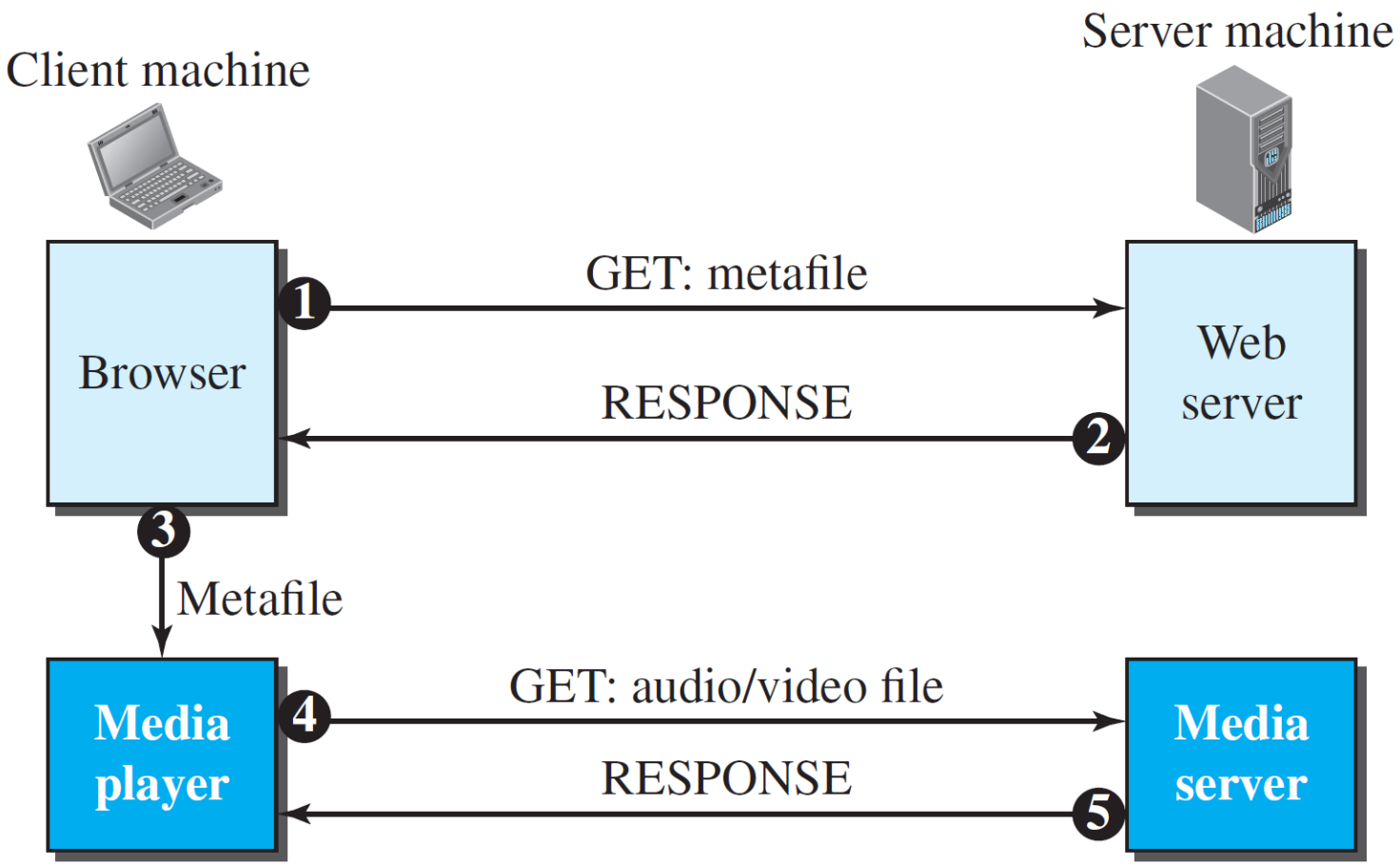
Using a Web Server with a Metafile



Using a Web Server with a Metafile

- The problem with the second approach is that the browser and the media player both use the services of HTTP. HTTP is designed to run over TCP.
- This is appropriate for retrieving the metafile, but not for retrieving the audio/video file.
- The reason is that TCP retransmits a lost or damaged segment, which is counter to the philosophy of streaming.
- We need to dismiss TCP and its error control; we need to use UDP.

Using a Media Server



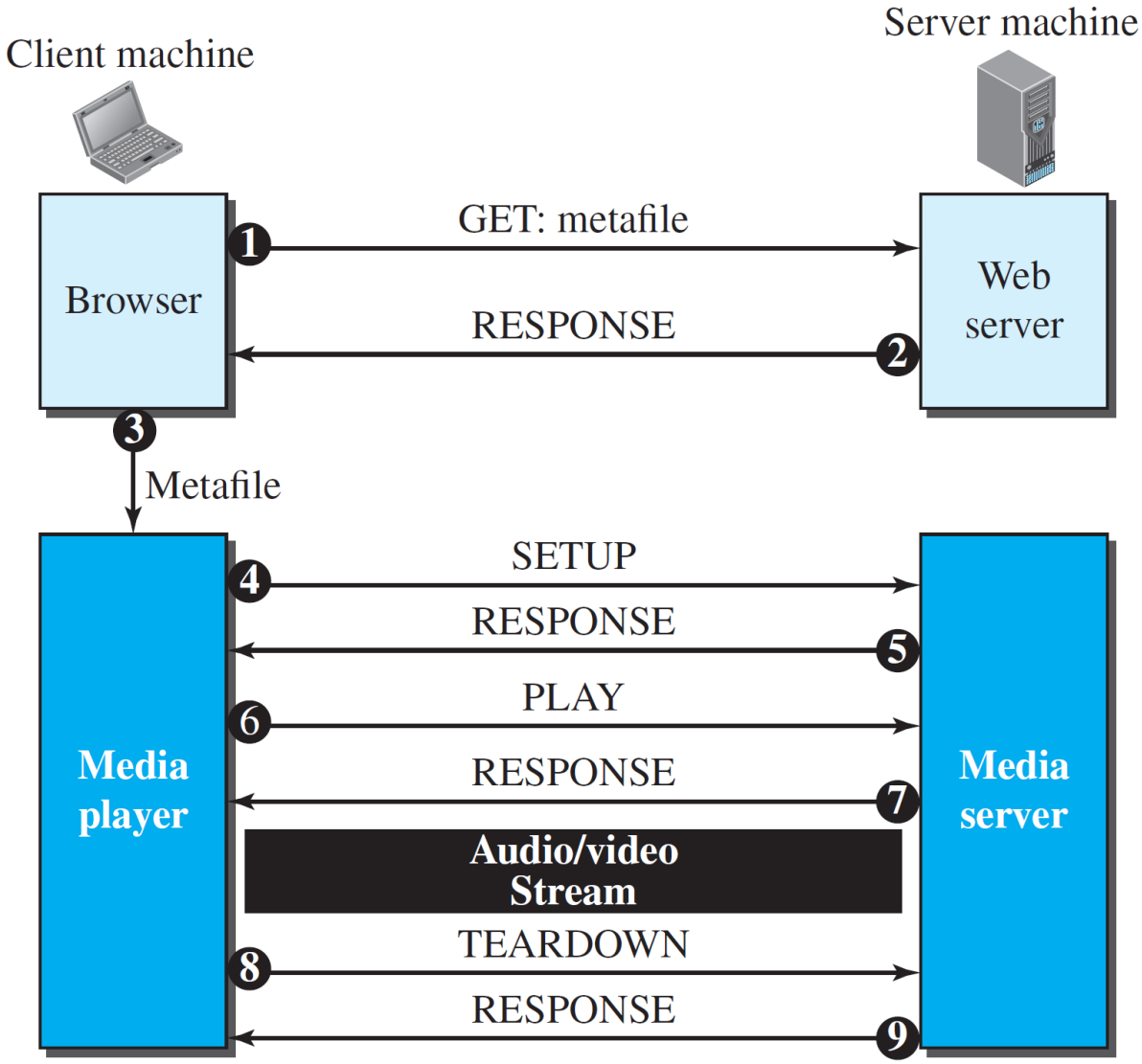
Using a Media Server and RTSP

- The **Real-Time Streaming Protocol (RTSP**) is a control protocol designed to add more **functionalities** to the **streaming** process.

- Using **RTSP**, we can **control** the playing of audio/video.

- RTSP is an **out-of-band** control protocol that is similar to the **second connection** in FTP.

Using a Media Server and RTSP



Streaming Live Audio/Video

– Example is **streaming** Internet **Radio** and Internet **TV**.

– There are **several similarities** between **streaming stored** audio/video and **streaming live** audio/video.

1. Both are **sensitive** to **delay**.

2. Both **cannot** accept **retransmission**.

Streaming Live Audio/Video

– However, there is a **difference**:

1. In **streaming stored** application, the communication is unicast and on-demand.

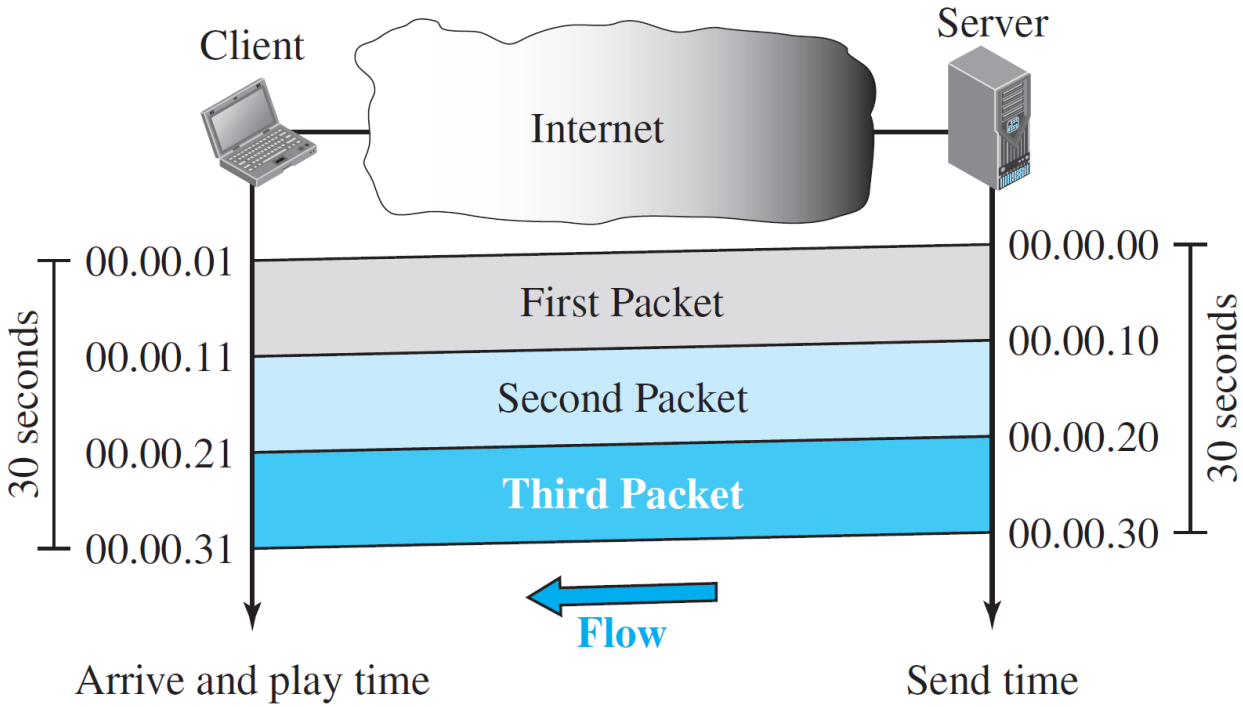
2. In **streaming live** , the communication is multicast and live.

Real-Time Interactive Audio/Video

- In this category people use the **Internet** to **interactively communicate** with one another.
- The **Internet phone** or **voice over IP** is an example of this type of application.
- **Video conferencing** is another example that allows people to communicate visually and orally.

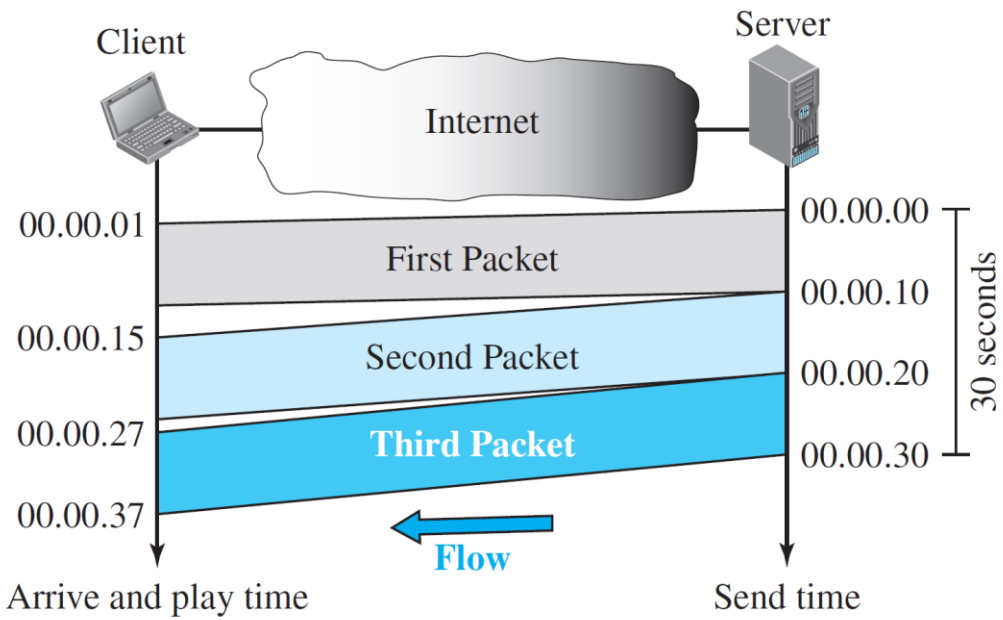
Characteristics

– ***Time Relationship***



Characteristics

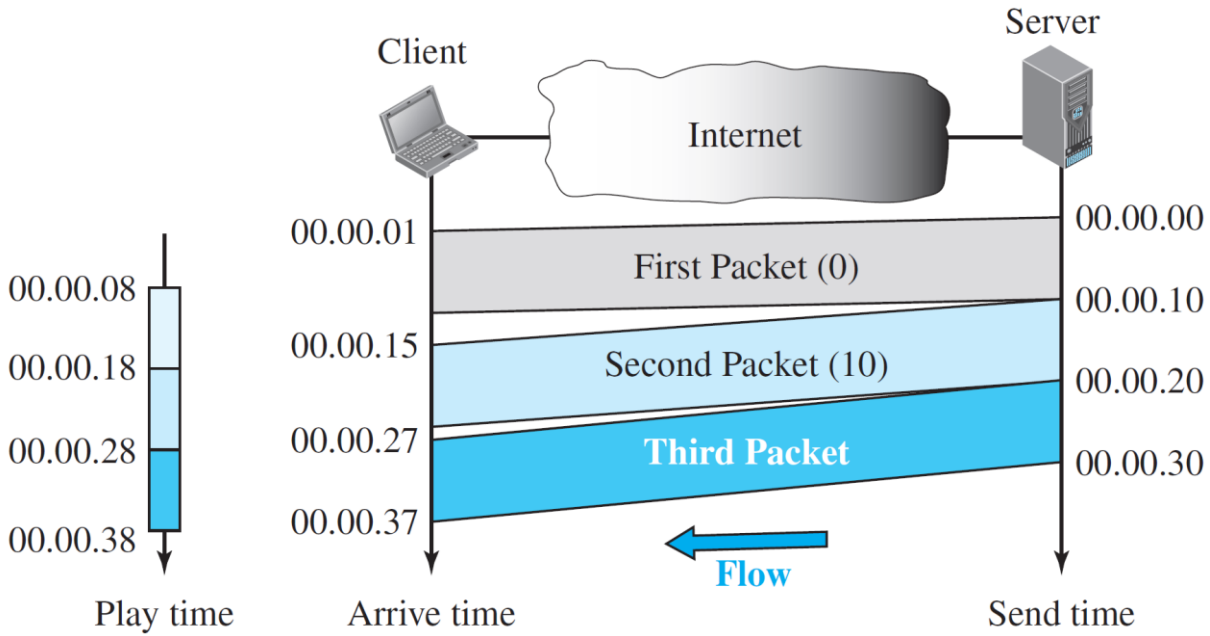
– ***Jitter in Time Relationship***



Characteristics

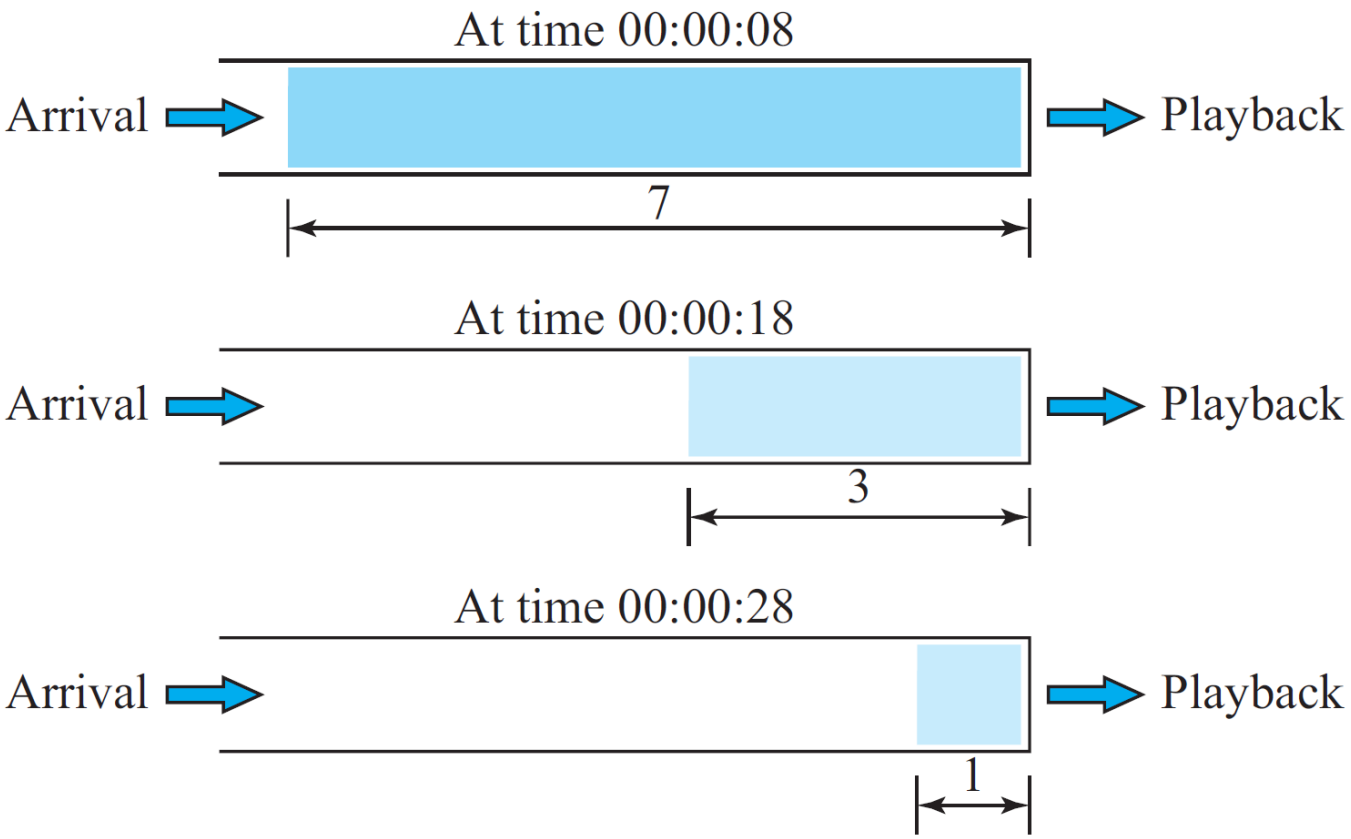
– ***Timestamp***

- To prevent jitter, we can **timestamp** the packets and **separate** the **arrival** time from the **playback** time.



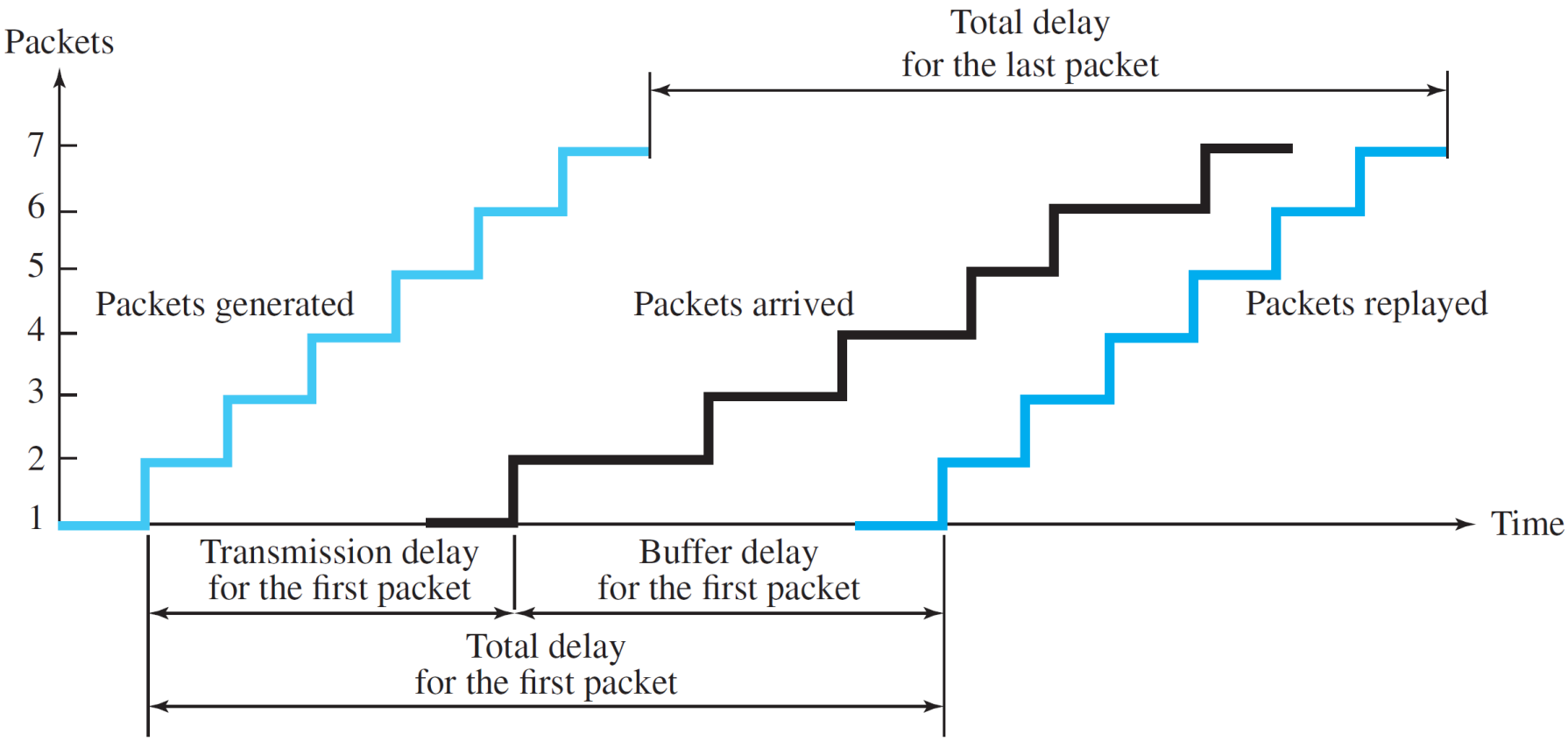
Characteristics

– ***Playback Buffer***



Characteristics

– *Playback Buffer – Time Line*



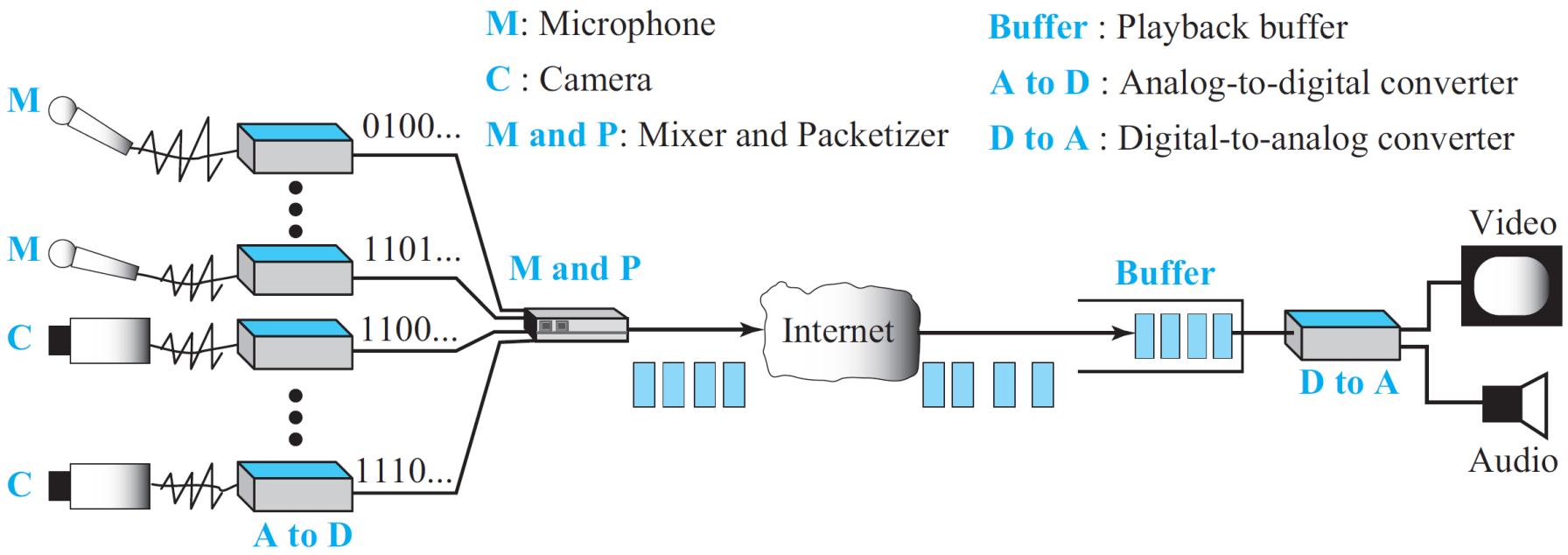
Example of a Real-Time Application: Skype

- Skype (**abbreviation** of the original project *Sky peer-to-peer*) is a **peer-to-peer** VoIP application software.

- The application allows registered users who have audio input and output devices on their PCs to make free **PC-to-PC** voice calls to other registered users over the Internet.

- Skype includes other popular features such as *instant messaging* (***IM***), *short message service* (***SMS***) group **chat**, file **transfer**, video **conferencing**, and **SkypeIn** and **SkypeOut** services.

Real-Time Multimedia System



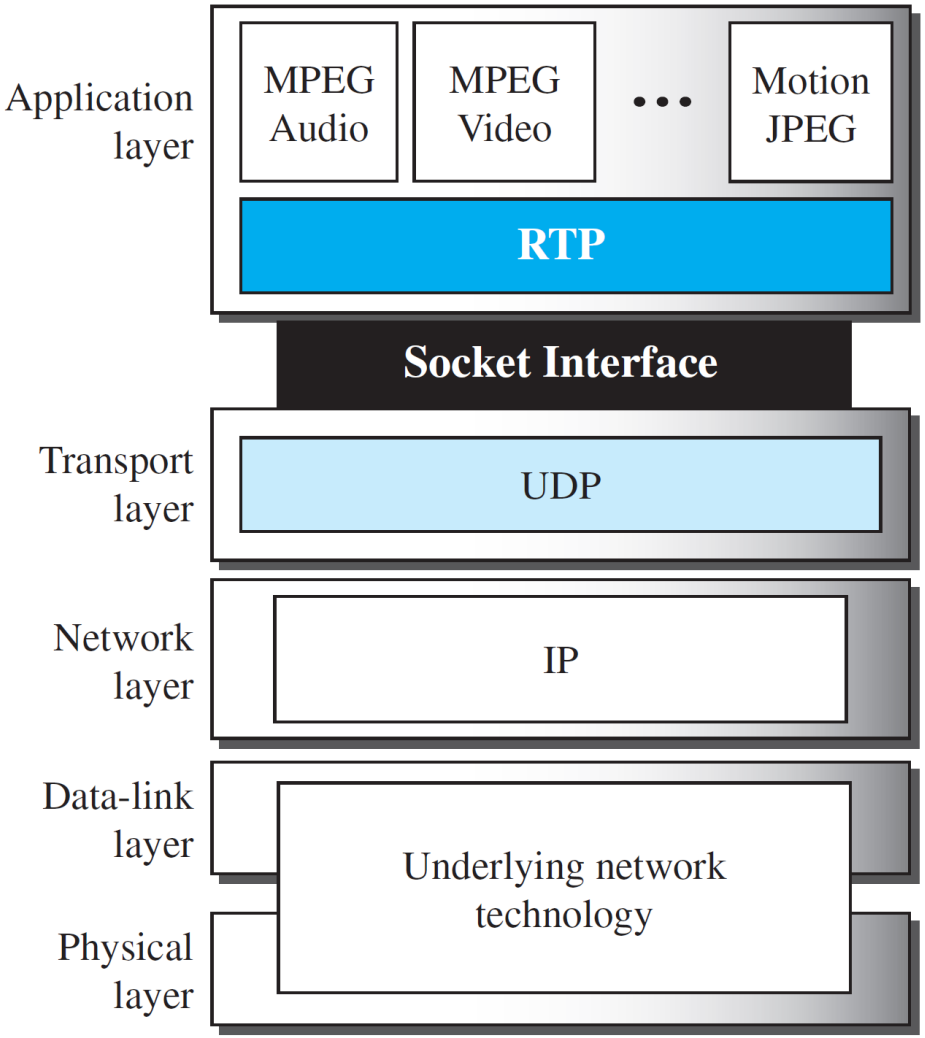
Real-time Transport Protocol (RTP)

- **Real-time Transport Protocol (RTP)** is the protocol designed to handle real-time traffic on the Internet.
- RTP is **used with UDP**.
- RTP stands between UDP and the multimedia application.

Real-time Transport Protocol (RTP)

- A socket interface is **located** between **RTP** and **UDP**.
- RTP is a **client-server** protocol.
- However, some **programming languages** provide some **facilities** to make the programming task easier.
- For **example**, the **C language** provides an RTP **library** and the Java language provides an RTP class.

RTP location in the TCP/IP protocol suite



Real-time Transport Control Protocol (RTCP)

- RTP **allows** only **one type** of **message**, one **that carries** data from the source to the destination.

- To really **control** the **session**, we need more communication between the participants in a session.

- **Control communication** in this case is assigned to a **separate** protocol **RTCP**.

Functions of RTCP

1. RTCP **informs** the sender of multimedia streams about the network **performance**, which is related to the **congestion** in the network.

– If the congestion is observed and reported by the RTCP, an application can:

- use a more aggressive compression method to reduce the number of packets and, therefore, to reduce congestion, for a trade-off in quality.

Functions of RTCP

1. Information carried in the **RTCP packets** can be used to **synchronize different streams** associated with the same source.

2. An RTCP packet can carry **extra information** about the sender that can be useful for the receiver, such as the name of the sender

Discussion and Questions